

Department of Computer Sciences	
Fundamentals of Programming & Data Science LAB	
Lab Number: 8	Topic: Introduction to Python Libraries NumPy, Pandas, Matplotlib & Basic Data Science Libraries
Teacher: Kanwal Khurshid	Session: Spring 2026
University of Engineering & Technology Lahore (Faisalabad Campus)	
Instructions for Students:	
Try to avoid code cheating and follow coding standards (Alignments & Comments where necessary). Pro Tip: (Create a word document with your registration# and divide each exercise into following steps to get highest scores) Problem Statement $\Rightarrow$ Code Snippet $\Rightarrow$ Output	

Objective

After completing this lab, students will be able to:

- Import and use Python libraries
- Work with arrays using NumPy
- Handle tables/data using Pandas
- Create charts using Matplotlib
- Perform simple calculations and visualization
- Understand basic data analysis workflow

Installing Libraries

Open terminal and run:

```
pip install numpy pandas matplotlib seaborn
```

Introduction to Libraries

Library	Purpose
NumPy	Numerical calculations and arrays
Pandas	Data handling and tables
Matplotlib	Graphs and charts
Seaborn	Beautiful statistical graphs

Practice:

Part1:Numpy:

Program 1 — Create NumPy Array

Program 2 — Array Mathematical Operations

Program 3 — Find Mean and Sum

Program 4 — Create Matrix

Program 5 — Random Numbers

PART 2 — PANDAS PRACTICE

What is Pandas?

Pandas is used for:

- Tables
- CSV files
- Data analysis

Program 6 — Create DataFrame

Program 7 — Display Specific Column

Program 8 — Add New Column

Program 9 — Find Maximum Marks

Program 10 — Read CSV File

PART 3 — MATPLOTLIB PRACTICE

What is Matplotlib?

Matplotlib is used for:

- Graphs
- Charts
- Visualization

Program 11 — Line Chart

Program 12 — Bar Chart

Program 13 — Pie Chart

Program 14 — Histogram

PART 4 — SEABORN PRACTICE

What is Seaborn?

Seaborn creates advanced and beautiful charts.

Program 15 — Seaborn Bar Plot

Program 16 — Seaborn Scatter Plot

STUDENT ASSIGNMENT TASKS (WITHOUT SOLUTION)

Assignment 1

Create a NumPy array of temperatures and find:

- Average temperature
- Highest temperature
- Lowest temperature

Assignment 2

Create a Pandas DataFrame for employee data:

- Name
- Salary

- Department

Display employees having salary greater than 50000.

Assignment 3

Read a CSV file containing student marks and display:

- Highest marks
- Lowest marks
- Average marks

Assignment 4

Create:

- Line chart
- Bar chart
- Pie chart

using any real-life dataset.

Assignment 5

Use Seaborn to create:

- Scatter plot
- Histogram
- Bar plot

Viva Questions

1. What is NumPy?
2. Difference between list and NumPy array?
3. What is Pandas used for?
4. What is a DataFrame?
5. Why do we use Matplotlib?
6. Difference between line chart and bar chart?

7. What is CSV file?
8. What is Seaborn?

BEST OF LUCK